SE2030 - Lab 02

Sprint Planning and Backlog Management for an Online Library System

Scenario: Online Library System

Project Overview

A university wants to develop a web-based Online Library System to support students and staff in accessing library services remotely. Currently, users must visit the physical library to search for books, check availability, or borrow items. The goal is to digitize core library functions, improve access to resources, and reduce manual workload for librarians.

Users of the System

- Students – Access library materials, borrow, reserve, and renew books.
- Academic Staff – Same as students, but with extended loan periods.
- Librarians – Manage inventory, track borrow/return activity, and configure system settings.
- Admin – Oversee users, reports, and system settings.

Functional Requirements (User Needs)

User Authentication & Roles

- Users should register/login using university credentials.
- System should assign roles (Student, Staff, Librarian, Admin) with appropriate permissions.

Book Search & Browsing

- Users can search for books by title, author, subject, ISBN.
- Filters for available, reserved, or overdue books.

Book Reservation

- Users can reserve a book if it's checked out.
- System notifies when the book becomes available.

Borrowing & Return Management

- Users can borrow and return books through the system.
- Borrowed books are automatically marked with due dates.
- Late returns are tracked with penalties.

Email Notifications

- Automated reminders for due dates, overdue fines, and reservation availability.

User Dashboard

- View borrowing history, current loans, due dates, and fines.

Librarian Dashboard

- Add/edit/delete book entries.
- Track inventory and current borrow status.
- Approve or manage reservations manually if needed.

Reporting

- Admins and librarians can generate reports (most borrowed books, user activity, overdue books, etc.).

System Settings

- Set loan durations, fine rates, library hours, and user roles.

Define the Scrum Roles

Product Owner

- University Librarian (or IT representative from the library)
- Prioritizes features based on student/staff needs.
- Maintains the product backlog.

Scrum Master

- IT Project Lead
- Facilitates Scrum events, protects the team from external interferences.

Development Team

- 4 developers (backend/frontend), 1 UI/UX designer, 1 QA tester

Prepare the Product Backlog

The product backlog includes a collection of user stories, which implies the services provided by the proposed web-based library system.

Sample Product Backlog

ID	User Story	Priority
PBI-01	As an authorized member, I want to log in securely so that I can access my account.	High
PBI-02	As a user (unregistered), I want to search for books by title, author, or subject.	High
PBI-03	As a student member, I want to reserve a book so that I can borrow it before it becomes unavailable.	Medium
PBI-04	As a librarian, I want to manage book inventory through an admin dashboard.	High
PBI-05	As a registered member, I want to receive email notifications for	Medium

	due dates.	
PBI-06	As a user, I want to view my borrowing history.	Low
PBI-07	As a Library Assistant, I want to reply to members' inquiries so that they get the required support related to library services.	Medium
PBI-08	As a Librarian, I want to approve new membership requests so that the system will forward the user credentials to new members.	High
PBI-09	As a user, I want to submit my membership application digitally so that I can obtain library membership efficiently.	High

Sprint Planning

The above sample requirements will be divided into two Sprints only for demonstration purposes.

1st Sprint – Week 4 to Week 7

Selected User Stories (according to their priorities): - ***Highlighted in yellow***

Sprint ID	User Story	Priority
SP1-01	As a user, I want to submit my membership application digitally so that I can obtain library membership efficiently.	High
SP1-02	As a Librarian, I want to approve new membership requests so that the system will forward the user credentials to new members.	High
SP1-03	As an authorized member, I want to log in securely so that I can access my account.	High
SP1-04	As a user (unregistered), I want to search for books by title, author, or subject.	High

Sprint Backlog

Task Breakdown for User Story 1 and Time Estimation (Membership Functionality)

Task ID	Task	Est. Hours
T-01.1	Design UI form for membership application	6
T-01.2	Create backend API to store membership requests	8
T-01.3	Validate user inputs (email, ID number, etc.)	3
T-01.4	Save application status as "Pending Approval"	2
T-01.5	Test membership application flow	3

Task Breakdown for User Story 2 and Time Estimation (Approve Membership Functionality)

Task ID	Task	Est. Hours
T-02.1	Design admin interface to view pending requests	5
T-02.2	Create backend endpoint to approve/reject requests	6
T-02.3	Auto-generate username/password upon approval	3
T-02.4	Send credentials to new members via email	4
T-02.5	Log approval action in system for audit	2
T-02.6	Test librarian approval flow	3

Task Breakdown for User Story 3 and Time Estimation (Login Functionality)

Task ID	Task	Est. Hours
T-03.1	Design login UI with fields for username and password	4
T-03.2	Implement secure authentication mechanism	6
T-03.3	Integrate login with university credentials system	5
T-03.4	Display login errors and success messages	2
T-03.5	Test login functionality	3

Task Breakdown for User Story 3 and Time Estimation (Book Search)

Task ID	Task	Est. Hours
T-04.1	Design search interface with filters	5
T-04.2	Implement backend search API	6
T-04.3	Enable search by title, author, subject, ISBN	4
T-04.4	Add filters for availability status	3
T-04.5	Test search feature with sample data	3

2nd Sprint – Week 8 – Week 11

Continue the same process for the rest of the prioritized user stories selected from the product backlog.